



Dental calculus reveals diet habits and medicinal plant use in the Early Medieval Italian population of Colonna

Angelo Gismondi^a, Alessia D'Agostino^a, Lorena Canuti^a, Gabriele Di Marco^a, Cristina Martínez-Labarga^a, Micaela Angle^b, Olga Rickards^a, Antonella Canini^{a,*}

^a Dipartimento di Biologia, Università di Roma "Tor Vergata", via della Ricerca Scientifica 1, Roma 00133, Italy

^b Soprintendenza per i Beni Archeologici del Lazio, Via Pompeo Magno 2, Roma, Italy

ARTICLE INFO

Keywords:

Middle ages
Food
Gas-chromatography mass-spectrometry
Plant microfossils
Starch granules
Chemical markers
Light microscopy

ABSTRACT

The community of Colonna (Rome, Central Italy), dated back to 8th–10th century CE, was characterized by a poor quality of life. In the present archaeobotanical investigation, we obtained a detailed qualitative reconstruction of the food habits of this Medieval population. To do it, light microscopy and GC–MS techniques were applied on the dental calculus of specimens exhumed from the cemetery of Colonna archaeological site, an approach never performed before on Italian Medieval samples. The identification of starch grains and other plant microfossils (e.g. pollen, phytoliths) showed a nutritional plan coherent with coeval poor classes, consisting of both C₃ (e.g. *Triticum* sp.) and C₄ (e.g. *Sorghum bicolor*) plant species, with a particular intake of acorns (Fagaceae) and legumes (Fabaceae) rich in proteins. Chromatographic analysis revealed the use of a large variety of foods: oil-rich seeds and fruits, aquatic resources (e.g. fish, molluscs, algae), animal derivatives (e.g. eggs, meat, milk, dairy products), Brassicaceae, Rosaceae fruits, aromatic herbs (Lamiaceae), mushrooms, wine and tea. In addition, specific markers of medicinal plants (e.g. *Digitalis* sp., *Hyssopus officinalis*, *Artemisia* sp., *Ephedra* sp.) evidenced the pharmaceutical culture of this historical period. Finally, the present work also amplified the knowledge about past environment, trade, plant uses and customs.

1. Introduction

The Middle Ages represents a long historical period characterized by deep political, religious, socio-economic and cultural changes which strongly influenced diet habits.

The current knowledge of Medieval foods essentially derives from ancient recipe books; however, such information is not fully exhaustive and should be integrated with other data reported in documents about breeding techniques, sale acts of lands, trades and archaeo-anthropological investigations.

In Medieval Italy, although several social realities existed, the common diet was mainly based on cereals and pulses, which were used to prepare breads or pottages. Sometimes, these foods were supplemented by other vegetables and, more rarely, by meat or fish, especially in wealthy classes. Meat was available to workers and peasantry but it was surely consumed in smaller quantities and lesser quality cuts than nobility (Giovannini, 1994; Montanari, 1979, 1988; Salvadori, 2003). Different cooking modalities were applied to meat; however, it was well known that boiling made this food sterile and much more digestible (Tuliani, 2001).

Milk and eggs were the most important protein sources for poor classes. In addition, dairy products were commonly employed and salted herring and some molluscs, being relatively cheap, represented accessible marine foodstuffs (Müldner and Richards, 2005). Other ordinary nourishments were collected from woods and forests, such as acorns, chestnuts, roots, mushrooms and wild fruits (Montanari and Brombert, 2015). Finally, wine consumption was predominant, not only for its energizing effect but also because water was often undrinkable and unhealthy (Piccinni, 2002).

Medieval daily life and even eating habits were subjected to the rigid rules that Church imposed on believers; it forced to eat fish instead of meat, use vegetable fats rather than animal ones and replace animal milk with almond juice, some days of the week and during the whole period of Lent (Montanari, 1988; Salamon et al., 2008; Tucci, 1985).

Detection and taxonomic identification of plant remains in ancient sites provide important ethnobotanical data. However, recently, several research groups demonstrated that the best way to study past diet is the direct analysis of the dental calculus found on human mandibular and maxillary rests (Cummings et al., 2016; Hardy et al., 2012; Henry et al., 2011; Radini et al., 2016). Indeed, mouth and teeth are the first organs

* Corresponding author.

E-mail address: canini@uniroma2.it (A. Canini).

to arrive in contact with food and the dental calculus, being a dense mineral deposit made up by inorganic salts and crystallized residues of oral microbiota, easily entraps in its matrix the organic molecules deriving from diet (Lindhe, 1984). Therefore, the dental calculus, persisting to archaeological contexts (Lieverse, 1999), represents a collection of compounds which are indisputable proofs of foods ingested during lifetime.

According to all this evidence, the main purpose of the present work was the determination, in qualitative terms, of ethnobotanical customs relative to a Medieval population who lived in Central Italy during the 8th–10th century CE. In particular, the investigations were carried out analysing the dental calculus of human remains found in the necropolis of Colonna (Rome, Latium). The original aspect of our research was the combined application of Light Microscopy (henceforth LM) and Gas-Chromatography Mass Spectrometry (henceforth GC–MS) on the ancient samples, approaches always performed separately in literature (Buckley et al., 2014; Cristiani et al., 2016; Cummings et al., 2016; Hardy et al., 2012; Henry et al., 2011). In conclusion, our study demonstrated a predominant usage of both Mediterranean and exotic plant species, as food and medicinal sources, in the Italian Early Medieval culture.

2. Materials and methods

2.1. Sample collection

Human remains found in the Medieval cemetery of the archaeological site of Colonna (Rome, Italy) were provided by the laboratory of Skeletal Biology and Forensic Anthropology (Dept. of Biology, University of Rome ‘Tor Vergata’) which performed the anthropological analyses of these samples (Baldoni et al., 2016). A total of 47 individuals, 34 adults and 13 sub-adults, were examined. Information about their estimated sex and age at death were obtained by Baldoni et al. (2016) and reported in Table 1. Between 1 and 30 mg of dark supragingival calculus per sample were collected and stored, at 4 °C, into Eppendorf tubes, until their analysis.

2.2. Sterilization and decontamination procedures

To decontaminate the whole working area from airborne modern plant material, detergents (starch-free soap), 5% sodium hypochlorite and UV light were employed to treat all surfaces, instruments, supplies and tools. First of all, fragments of masticatory apparatus were gently dry-brushed with a single-use toothbrush. After dental calculus collection, in order to eliminate any past and modern environmental contaminants from calculus surface, each sample was UV-treated for 10 min on each side, soaked in 5% sodium hypochlorite for 15 min, washed twice in sterile bidistilled water (40 °C), rinsed in 100% ethanol, to remove the aqueous component, and let it dry, at 37 °C, in a new sterile Eppendorf tube. All these steps were obtained by joining several literature guidelines (Cristiani et al., 2016; Crowther, 2012; Cummings et al., 2016; Gismondi et al., 2016; Hardy et al., 2016; Henry et al., 2014; Weyrich et al., 2017). In addition, all procedures were carried out under a sterile vertical laminar flow hood (Heraeus HER-Asafe HS12 Type). To validate the sterilization and decontamination protocols, five dental calculus were randomly selected and subjected to the following analysis. In particular, samples were washed with 200 µL of water which was partially (100 µL) observed by light microscopy. Only rare plant microremains, such as fungal spores, were detected. Then, the same samples were treated according to our previous cleaning protocols and washed again with 200 µL of water. In this last case, the microscopic investigation of the washing solution did never show any plant microremains, demonstrating that these procedures were necessary to remove all external impurities. Similarly, these controls were also performed by GC–MS. Indeed, the remaining 100 µL of both washings were completely dried out, resuspended in 100%

Table 1

Biological profile of the Medieval specimens. All individuals exhumed from Colonna necropolis were associated to a code, reporting tomb (T) and stratigraphic unit (US) number, according to Baldoni et al. (2016). For each specimen, estimated sex (F, female; M, male; NDA, adults with not determined sex; NDS, sub-adults with not determined sex because of their sexual immaturity), estimated age at death (GA, generic adult individual; GS, generic sub-adult individual) and district of the masticatory apparatus used for dental calculus sampling were reported. In particular, sampled teeth were codified according to Universal Teeth Numbering System (American Dental Association, 1999).

Samples	Sex	Age (years)	Calculus location	
			Teeth	Surface
4	F	36–50	11,12,26,27	Lingual
6 US 63	NDS	2–3	S,T	Buccal
8 US 64	F	36–50	24,25	Lingual
12 US 74	NDS	5–8	S	Buccal
13 US 73	NDS	3–9	K	Lingual
14 US 75	M	36–50	19	Buccal
18 US 79	M	36–50	3,4,25,26	Lingual
19 US 81	M	18–35	20,21,22	Lingual
22 US 84	F	36–50	14,21,22	Buccal
25 US 87	F	18–35	23,24,25,26	Lingual
26 US 88 1A	F	30–35	7,8	Lingual
27 US 89	F	18–35	14,26	Buccal
30 US 95	F	36–50	2,3,19	Buccal and lingual
31 US 96	F	18–35	2,17,18,19	Buccal
32 US 97 1A	M	20–25	24,25	Lingual
33 US 98	NDA	18–25	2,5,11,17,18,19,22	Buccal
37 US 104	M	23–25	13,24,25	Lingual
39 US 106 A	M	15–18	4,10,30,31	Buccal and lingual
40 US 107 A	NDA	33–55	31,32	Buccal and lingual
40 US 107 B	NDA	GA	16,32	Buccal and lingual
42 US 109	NDS	GS	M,30	Lingual
44 US 114	NDS	12	14,27	Lingual
48 US 124	F	18–35	2,30,31	Buccal
49 US 125	M	35–50	13,14,25,26	Lingual
51 US 127	M	36–50	4,5,19	Buccal
52 US 134	NDS	10–12	8,25	Lingual
53 US 135	F	36–50	6,7,27	Lingual
54 US 140	NDS	13–16	3	Buccal
57 US 152	NDS	5–7	L,T	Buccal
63 US 185	F	36–50	19	Buccal and lingual
65 US 186	F	36–50	5,7,21,22	Lingual
67 US 188	NDS	14–16	7,8,9,30,31	Buccal and lingual
68 US 200	NDS	GS	19,20	Buccal and lingual
69 US 204	M	36–50	12,13,30,31,32	Lingual
71 US 216	F	25–35	18,19,20	Lingual
71 US 247	M	36–50	5,6,18,21	Buccal
80 US 354 1E	NDS	8–10	3	Lingual
80 US 354 2D	NDS	7–9	I,J	Buccal and lingual
80 US 354 3A	F	36–50	8,27,30	Lingual
80 US 354 4B	NDA	36–50	8,10,11,21	Buccal and lingual
80 US 354 5C	M	35–50	9,10,28,30,31	Buccal and lingual
81 US 247 1A	M	36–50	12,13,14,15,18,20,21	Buccal
83 US 73 A	M	35–50	26,27	Lingual
83 US 179 1A	M	30–40	18,20,30	Buccal
83 US 179 B	F	36–50	18,19	Buccal and lingual
83 US 179 C	NDS	14–17	3,15	Buccal and lingual
84 US 211	F	18–20	14,30	Buccal

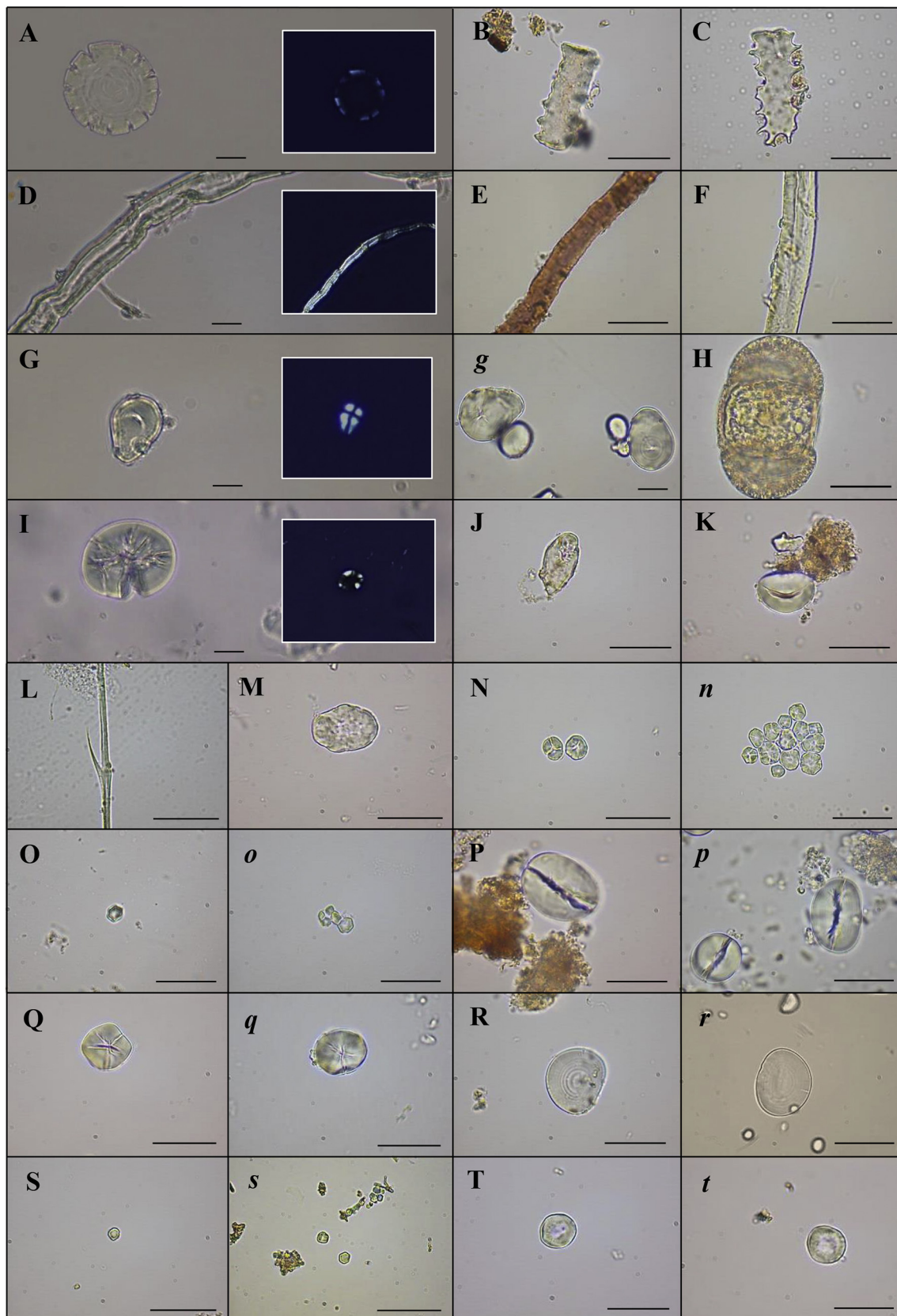
hexane and treated and analysed as reported in the relative paragraph (GC–MS analysis). No organic molecule was revealed after the decontamination.

2.3. LM analysis

To extract starch granules and other microremains from the dental calculus, a protocol based on Hardy et al. (2009) method was used, after appropriate modifications. From 1 to 20 mg of sample, where available, were resuspended with 500 µL of 1 M HCl, sonicated for 10 min (Falc Instruments MOD: LBS1 34) and left in agitation for 24 h. After centrifugation at 13.000 rpm for 10 min, the pellet was subjected

to 3 consecutive washings with bidistilled water. The last pellet was resuspended in 100 μ L of bidistilled water and glycerol (1:1), sonicated for 5 min and examined at LM (Nikon Eclipse E100). Each starch grain,

pollen, calcium oxalate crystal, plant fibre and phytolith was photographed at 100 $\times$ magnification, using a program for capturing images (ProgRes CapturePro 2.9.0.1), and measured in size by SuperAmpelo



(caption on next page)

Fig. 1. Plant microremains at light microscopy. Examples of plant microremains found in the dental calculus of the samples were shown: *Triticum* sp. starch granule and relative polarized image (A); Poaceae phytoliths (B and C); plant fibre and relative polarized image (D); plant fibres (E and F); *Q. ilex* starch granule, relative polarized image (G) and relative modern experimental reference (g); *Pinus* sp. pollen grain (H); Fabaceae starch granule and relative polarized image (I); Fabaceae starch granule (J); Fabaceae starch granule (K); fragment of feather barbule (L); unidentified starch granule (M); *P. miliaceum* starch granules (N) and relative modern experimental reference (n); *Avena* sp. starch granule (O) and relative modern experimental reference (o); *C. arietinum* starch granule (P) and relative modern experimental reference (p); *S. bicolor* starch granule (Q) and relative modern experimental reference (q); *T. dicoccum* starch granule (R) and relative modern experimental reference (r); *P. nigrum* starch grain (S) and relative modern experimental reference (s); *D. carota* starch grain (T) and relative modern experimental reference (t). The black bar indicates 10 μ m.

2.0 software. Taxonomic identification of microremains at species, genus or family level was performed by direct comparison of their sizes, structural characteristics (e.g. form, aggregation level) and ultra-structural elements (e.g. lamellae, hilum) with literature data, including a pollen Atlas (Travaglini et al., 2014), and a starch reference collection present in our laboratory and published in Gismondi et al. (2018a). In detail, this library, hosted at the Department of Biology – University of Rome “Tor Vergata”, includes starch granules belonging to 40 modern plant species grown in the Botanical Garden of Rome “Tor Vergata” or purchased by local certified sellers. Experimental collection species were selected for their original geographic distribution area (e.g. Mediterranean basin) and dietary use. Beyond the most common starchy species, such as cereals and legumes, we also investigated starches of poor and cheap plant foods (e.g. acorns, tubers), widely consumed in the Middle Ages, to provide more adequate archaeological interpretations. Starch reference collection was obtained analysing a total of 900 randomly selected starch granules per species.

2.4. GC–MS analysis

A minimum amount of 10 mg of dental calculus for sample were considered necessary to obtain significant results. The mineral matrix was resuspended in 1 mL of 6% HCl for three days, in agitation. Once solubilized, 1 mL of hexane was added and sample was vortexed for 2 h. After centrifugation at 10.000 rpm for 10 min, the supernatant hexane fraction was recovered and dried out by a speed-vac system (Eppendorf AG 22331 Hamburg, Concentration Plus). The pellet was derivatized by resuspension with 50 μ L of hexane and 50 μ L of the Methyl-8-Reagent (Thermo Scientific), according to manufacturer's guidelines. Two microliters of extract were injected in a GC–MS system (QP2010, Shimadzu, Japan) and, at least, 3 runs for each sample were carried out. The sample was separated by a DB-5 Phenomenex capillary column (length 30 m $\times$ diameter 0.25 mm $\times$ thickness 0.25 μ m) using the following method: initial oven temperature at 60 $^{\circ}$ C for 5 min; then, at a rate of 6 $^{\circ}$ C/min, the oven was set at 150 $^{\circ}$ C for 5 min, at 250 $^{\circ}$ C for 5 min and, finally, at 330 $^{\circ}$ C for 25 min. Helium was used as carrier gas at a constant flow of 1 mL/min. Mass spectrum of each compound was obtained by electron impact (EI) at 70 eV (scanning from 100 to 600 m/z), ion source temperature 230 $^{\circ}$ C, interface temperature 320 $^{\circ}$ C and solvent cut time of 6 min. Each molecule was identified by comparing their mass profiles to those of NIST (National Institute of Standard and Technology) Library 14 loaded in detection software of the instrument (Solution software). Similarity values were considered acceptable only if higher than 85%. The analysis was carried out in qualitative and non-quantitative terms, as already performed in Gismondi et al. (2018b).

3. Results

Ethnobotanical traditions of the ancient inhabitants of Colonna were investigated, identifying the plant species used for both dietary and medical purposes.

In the Medieval cemetery, 144 individuals were exhumed but only 47 of them showed dental calculus (Table 1). After sampling, all potential environmental contaminants were rigorously removed from calculus surface, adopting accurate protocols and meticulous controls, as reported in Materials and methods section.

LM analysis detected microremains in 40 individuals (Fig. 1). In total, 104 starch granules, 6 not determined Poaceae phytoliths, 3 calcium oxalate crystals, 12 fragments of plant fibres, 2 pollen grains and 1 fragment of bird feather were found in the calculus samples (Table 2).

Nine individuals did not show starches, while, in some cases, single specimens simultaneously revealed the presence of different types of starch. All ancient starches were clustered in 12 morphotypes and each one of them was described, by morphologic and morphometric parameters (Table 3). The taxonomic level of morphotypes was identified, comparing them with modern collection (Gismondi et al., 2018a). To support our interpretations, the features of experimental references were also added in Table 3.

At population level, starch grains of Fabaceae resulted to be dominant; some of them were specifically attributed to *Quercus ilex* L. species, an oak typical of Mediterranean scrub whose seeds might be used as food. Seven Fabaceae starch granules were observed but only one of them could be associated to *Cicer arietinum* L. Starches belonging to cereals, both with C₃ and C₄ photosynthetic cycle, were also detected: *Sorghum bicolor* L. Moench, *Triticum* sp., *Panicum miliaceum* L., *Triticum dicoccum* L. and *Avena* sp. Calcium oxalate crystals, plant fibres and phytoliths confirmed the employment of Poaceae and herbaceous plants. Finally, two starches of *Daucus carota* L. and three of *Piper nigrum* L. were found. The high number of undetermined starch grains was related to degradation phenomena due to Ptilin enzymatic activity (an α -amylase, secreted by salivary glands, which performs the first digestion step of carbohydrates in mouth), grinding procedures and cooking processes, especially if performed in presence of water.

The detection of *Pinus* sp. and *Juniperus communis* L. pollen grains sustained the presence of these Gymnosperms in the area.

A fragment of feather barbule with triangular-shaped nodes, attributable to Anseriform birds, according to literature reference (Cristiani et al., 2016; Dove and Koch, 2011), hypothesized ingestion or inhalation of this microremain due to plucking activity of waterfowls.

GC–MS was performed only on the specimens who presented enough dental calculus (21 individuals) (Fig. 2). The analysis was repeated, for each sample, at least 3 times and, in all cases, it always showed similar profiles. In Supplementary material 1, the molecules identified in each specimen were clustered in biochemical classes and listed. To rough out the diet of this community, specific foods (e.g. mushrooms of *Amanita* genus, Persoon) or food categories (e.g. Brassicaceae), ingested in life at least once, were deduced by studying and associating of the detected compounds, using literature data and scientific food databases (FoodB, 2013; TGSC, 2015).

All chromatograms showed a dominant presence of C₄ to C₃₅ *n*-alkanes and *n*-alkenes which represented the degradation forms of both oral microbiota and animal or plant food molecules. However, Eglinton et al. (1962) and Buckley et al. (1999) proposed that such type of organic substances specifically derived from the decomposition of unsaturated or saturated plant oils, suggesting the use of resins, waxes, fruits and vegetables. However, being abundant and aspecific, these compounds were not inserted in Supplementary material 1.

In each sample, we recovered methyl esters of saturated fatty acids, such as decanoic (C_{10:0}), capric (C_{14:0}), pentadecanoic (C_{15:0}), palmitic (C_{16:0}) and stearic (C_{18:0}). The detection of these molecules together with unsaturated fatty acids, including oleic (C_{18:1}), linoleic (C_{18:2}),

Table 2

Number of identified plant microremains. For each Medieval specimen, the number of starch granules identified in the dental calculus by light microscopy analysis and associated to a plant species, genus or family was indicated (the amount of unidentified starch grains was also shown). In addition, the detection of other microremains (F: plant fibres; P: Poaceae phytoliths; O: calcium oxalate crystals; G: pollen grains; A: fragment of feather barbule) was also reported. The total number of microremains was shown, for each individual and every taxon. Fab: Fabaceae; Fag: Fagaceae; Poa: Poaceae; ND: Not determined; Tot: Total; OM: other microremains.

Sample name	<i>Avena</i> sp.	<i>Cicer arietinum</i>	<i>Daucus carota</i>	<i>Panicum miliaceum</i>	<i>Piper nigrum</i>	<i>Quercus ilex</i>	<i>Sorghum bicolor</i>	<i>Triticum</i> sp.	<i>Triticum dicoccum</i>	Fab	Fag	Poa	ND	Tot	OM
4						1					1		2	4	
6 US 63							1						1	2	
8 US 64														0	
12 US 74													1	1	
13 US 73														0	
14 US 75													1	1	
18 US 79									1		3		3	7	
19 US 81										1			1	2	1F
22 US 84					1	1				1			7	10	
25 US 87					1					1				2	
26 US 88 1A						1				2			4	7	1F
27 US 89													1	1	2P
30 US 95														0	
31 US 96														0	1O
32 US 97 1A										1		1		2	1G
33 US 98											2		1	3	1F
37 US 104														3	
39 US 106 A							1			1			1	3	
40 US 107 A				2						1			2	5	2F
40 US 107 B														4	
42 US 109													1	1	
44 US 114									1					1	
48 US 124													2	2	1P
49 US 125							1			1				1	3
51 US 127													2	2	
52 US 134														0	
53 US 135										1	1		1	3	
54 US 140											1			1	
57 US 152						1							1	2	1O
63 US 185		1	1								1		1	4	1P
65 US 186							1						1	2	1A
67 US 188			1			1								2	2P
68 US 200													1	1	
69 US 204								1		1	2			4	2F
71 US 216							1							1	1F
71 US 247	1				1					1				3	1F
80 US 354 1E													1	1	
80 US 354 2D						1	1			1			1	4	1O
80 US 354 3A								1						1	
80 US 354 4B														0	
80 US 354 5C														0	1G
81 US 247 1A							1				1			2	1F
83 US 73 A				1				1					1	3	
83 US 179 1A													2	2	1F
83 US 179 B														0	
83 US 179 C														0	
84 US 211													2	2	1F
Total	1	1	2	3	3	6	7	3	2	6	19	1	50	104	24

eicosadienoic (C_{20:2}) and eicosatrienoic (C_{20:3}), indicated the intake of plant oils (e.g. contained in acorns) and animal fats (Di Fiore and Chianese, 2008; Kanthilatha et al., 2014; Rakić et al., 2006). In particular, the detection of unsaturated fatty acids could be linked to oil-rich seeds (e.g. pine nuts) and fruits (e.g. olives).

Derivatives of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), polyunsaturated omega-3 fatty acids, mainly ascribable to aquatic sources (e.g. molluscs, blue fish, algae, fish-oil) and a lesser extent to dried fruits (e.g. nuts, almonds, sunflower) (Kris-Etherton et al., 2003; Swanson et al., 2012), were highlighted in 9 specimens.

The association of glucosinolate derivatives (e.g. glucobrassicin, isothiocyanic acid methyl ester and isothiocyanatoacetaldehyde), erucic acid (C_{22:1}) and tetracosenoic acid (C_{24:1}), found in the major part of the individuals, suggested the ingestion of Brassicaceae (e.g. cabbage, cauliflower).

A few cholesterol derivatives and D-galactose molecules testified the intake of animal protein foodstuffs: meat, eggs, milk and dairy products, maybe of sheep, goat or donkey (Ginatempo, 1984).

One-half of the community showed various lactones (γ-octalactone, γ-pentalactone, γ-caprolactone) which strongly suggested the use of fruits belonging to the Rosaceae family, including plums, peaches, black cherries and quinces.

Our attention was captured by the identification of tartaric acid in the ancient dental calculus of two individuals. Indeed, this compound, abundant in grapes, might be correlated with wine drinking (Badler et al., 1990; Guasch-Jané et al., 2004; Salvini et al., 2008). The detection of several mono/sesquiterpene alcohols (e.g. limonene oxide, linalool, +/–lavandulol, S-dihydrocitronellol, alpha-ocimene, cumene hydroperoxide, bergamiol and nerolidol) sustained the putative application of Lamiaceae herbs.

Table 3

Starch grain morphotypes. Starch morphotypes identified from ancient dental calculus and their relative experimental reference (ER; extrapolated by [Gismondi et al., 2018a](#)) were described.

Morphotype	Taxonomic group	Morphologic and morphometric description
I	<i>Avena</i> sp.	Polyhedral shape; granule size was 2.83 µm in length and 2.05 µm in width; centric hilum.
ER I	<i>Avena</i> sp.	Polyhedral shape; size range: 2–9 µm in length and 2–9 µm in width; indistinct lamellae; unfissured granules; centric hilum.
II	<i>Cicer arietinum</i>	Ovoid shape; granule size was 14.29 µm in length and 11.23 µm in width; lamellae are slightly evident as close concentric rings; presence of a very pronounced longitudinal centric fissure.
ER II	<i>C. arietinum</i>	Ovoid to ellipsoid shape; size range: 5–25 µm in length and 3–15 µm in width; eccentric, complete and distinct lamellae; mesial longitudinal cleft; obscured hilum.
III	<i>Daucus carota</i>	Grains were disk shaped; size range: 5–9 µm in length and 4–5 µm in width; presence of feeble multiple radial fractures (starting from the amorphous convex centre of the structure); slightly visible lamellae as close concentric rings; hilum slightly visible.
ER III	<i>D. carota</i>	Spherical to ovoid shape; size range: 3–38 µm in length and 3–25 µm in width; concentric, incomplete and indistinct lamellae; radial fissures; centric and distinct hilum in some granules.
IV	<i>Panicum miliaceum</i>	Polyhedral grains with blunted edges; size range: 2.5–4 µm both in length and in width; centric hilum; close concentric slightly detectable lamellae.
ER IV	<i>P. miliaceum</i>	Polyhedral (rounded off) shape; size range: 1–7 µm in length and 1–7 µm in width; concentric and indistinct lamellae; unfissured granules; centric distinct hilum.
V	<i>Piper nigrum</i>	Polyhedral granules with pentagonal or hexagonal concave faces and acute edges; size range: 1–2 µm both in length and in width; typical bright boundary.
ER V	<i>P. nigrum</i>	Polyhedral (sharp-edged) shape; size range: 1–4 µm in length and 1–4 µm in width; indistinct lamellae; unfissured granules; centric and indistinct hilum.
VI	<i>Quercus ilex</i>	Grains were drop-shaped; size range: 5–8 µm in length and 4–6 µm in width; hilum not clearly evident; presence of the typical pore at the narrow end.
ER VI	<i>Q. ilex</i>	Ovoid to pyriform shape; size range: 6–25 µm in length and 5–10 µm in width; eccentric, incomplete and indistinct lamellae; radiate fissures; eccentric and indistinct hilum; pore at one end.
VII	<i>Sorghum bicolor</i>	Rounded grains with some peculiar flattened surfaces; size range: 7–11 µm in length and 6–10 µm in width; deep radial cracks starting from a centric hilum.
ER VII	<i>S. bicolor</i>	Ovoidal shape; size range: 2–20 µm in length and 2–15 µm in width; indistinct lamellae; radiate fissures; centric obscured hilum.
VIII	<i>Triticum</i> sp.	Grains were disc-shaped; size range: 4–9 µm in length and width; hilum not evident; concentric lamellae were present.
ER VIII	<i>Triticum</i> sp.	Ovoid to discoidal shape; size range: 1.5–35 µm in length and 1.5–25 µm in width; concentric or eccentric, complete and distinct lamellae; unfissured granules; centric and indistinct hilum.
IX	<i>Triticum dicoccum</i>	Grains were irregular disc-shaped; size range: 6–30 µm in length and 5–20 µm in width; clear concentric kidney-shaped lamellae were distributed on the whole grain.
ER IX	<i>T. dicoccum</i>	Ovoid to discoidal shape; size range: 2–30 µm in length and 2–20 µm in width; eccentric, complete and distinct lamellae; unfissured granules; centric indistinct hilum.
X	Fabaceae	Irregular reniform grains; size range: 6–19 µm in length and 3–14 µm in width; hilum is not detectable; quite clear concentric lamellae; presence of a longitudinal crack in the amorphous central area.
ER X	Fabaceae	Ovoid to ellipsoid to reniform shape; size range: 4–40 µm in length and 3–28 µm in width; concentric, complete and distinct or indistinct lamellae; mesial longitudinal cleft; obscured hilum.
XI	Fagaceae	Oblong or drop-shaped grains; size range: 5–15 µm in length and 3–11 µm in width; faintly visible lamellae; presence of longitudinal fractures.
ER XI	Fagaceae	Ovoid to pyriform shape; size range: 2–25 µm in length and 1–15 µm in width; indistinct lamellae; longitudinal fissured or unfissured granules; eccentric indistinct or invisible hilum.
XII	Poaceae	Small irregular microremain whose morphotype was ascribable to various Poaceae starch grains; granule size was 3.70 µm in length and 3.90 µm in width; presence of concentric lamellae.
ER XII	Poaceae	Small starch granules of all Poaceae species present a spherical form whose range is 1–4 µm in length and width; concentric lamellae; centric and indistinct hilum.

GC–MS analysis also revealed the presence of plant secondary metabolites ascribable to medicinal species. For instance, theophylline, an alkaloid synthesized in tea plant (*Camellia sinensis* L. Kuntze) was found in 4 specimens. Moreover, the calculus of an adolescent of 14–16 year old (T67 US188) presented muscimol, a poisonous isoxazolic alkaloid whose production occurs in the fungal genus of *Amanita* ([Hall and Hall, 1994](#)).

Finally, the following peculiar metabolites were also identified: i) a cardenolide (detected in T25 US87), usually contained in *Digitalis* sp. and Apocynaceae family; ii) the isopinocampheol (in T51 US127), a terpenic alcohol of *H. officinalis* L.; iii) isoforms of artemisia ketone and alcohol (T18 US79; T52 US134; T69 US204; T71 US247) which are typical markers of *Artemisia* sp. (Asteraceae family); iv) ephedrine derivatives (T4; T18 US79; T39 US106A; T52 US134; T67 US188; T80 US354 4B; T83 US179 1A) produced in *Ephedra* sp.

General cooking practices of foods were proved by revealing indicators of burned charcoal, such as naphthalene and toluene ([Buckley et al., 2014](#)).

4. Discussion

Human diet reconstruction reveals uses, customs, culture and socio-economic aspects of ancient civilizations but also ecological and environmental characteristics of past habitats. Therefore, for the first time

in literature, food habits of an Early Medieval population were extrapolated by studying deposits of dental calculus.

In [Baldoni et al. \(2016\)](#), the Medieval population of Colonna was reported to be a poor community of farmers and animal breeders which were subjected to strenuous activities and precarious sanitary conditions. In general, isotopic data revealed, for these individuals, a diet essentially based on terrestrial foods, both C₃ plants and animal resources.

The relevance of our work consisted in the combined application of two different analytical techniques, light microscopy and gas-chromatographic mass-spectrometry, on Medieval dental calculus. Indeed, both these approaches can produce very interesting and important qualitative data. LM approach, detecting starch granules embedded in calculus flakes, provided direct indication about the main carbohydrate sources introduced by diet. Moreover, the microscopic identification of pollens and phytoliths gives further information on plant use. On the other hand, GC–MS highlights, in the mineral deposit, the existence of specific (e.g. alkaloids) or generic (e.g. fatty acids) markers belonging to foods consumed, at least once, in lifetime ([Buckley et al., 2014](#); [Hardy et al., 2012](#); [Henry et al., 2011](#)). It is evident that the number of starch granules and/or molecules found in the calculus of each individual cannot be considered a quantitative index of the usage of the corresponding food but, at population level, it represents a direct proof of the consumption of these plant species ([Leonard et al., 2015](#)).

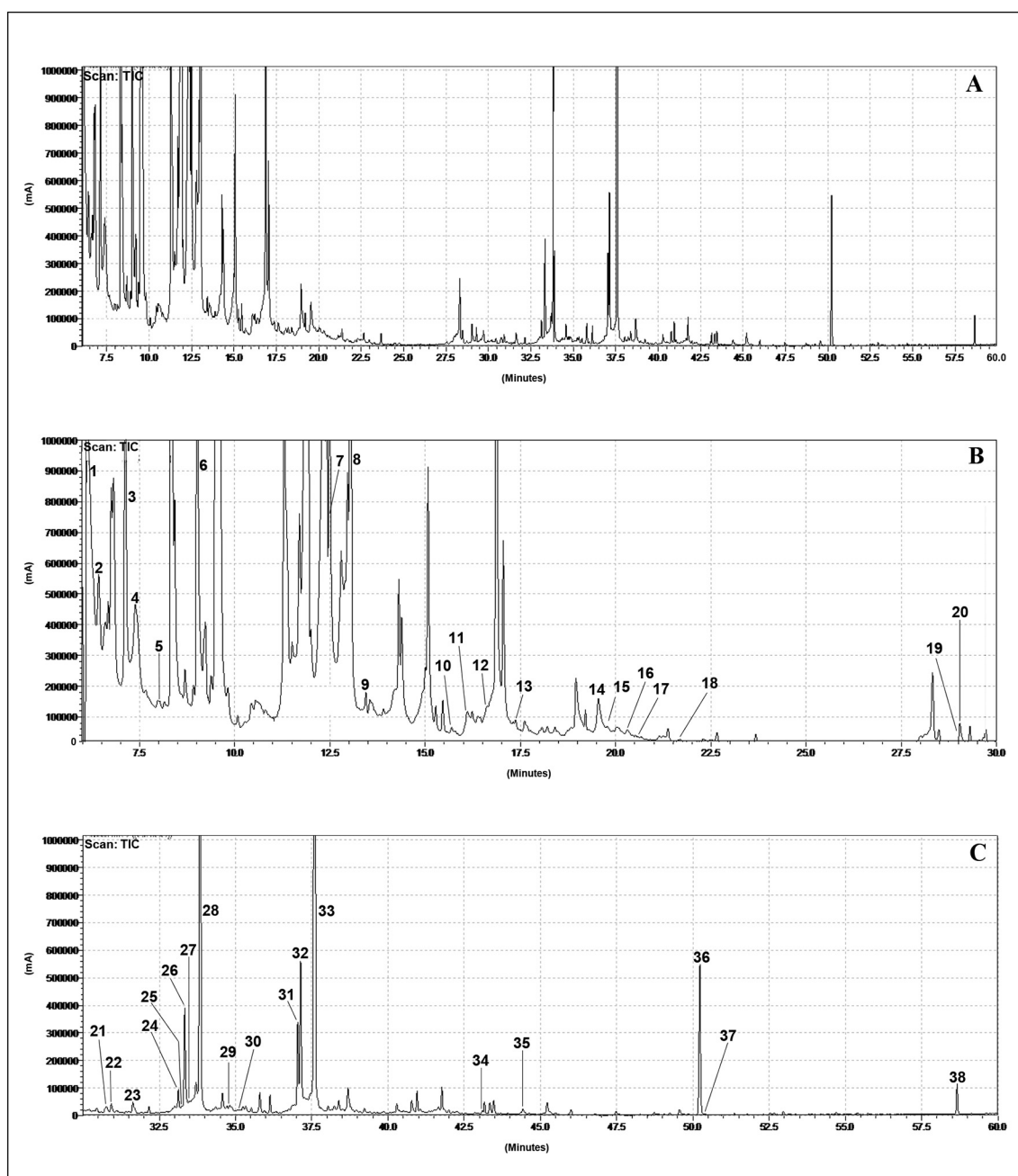


Fig. 2. GC–MS chromatogram. A representative GC–MS profile was shown (T67 US188) (A panel). The chromatogram was split in two parts (0–30 min, B panel; 30–60 min, C panel) and magnified. Peaks relative to molecules identified in the dental calculus of this specimen (Supplementary material 1, T67 US188 sample) and used for its diet reconstruction were indicated (1: α -D-Galactopyranose, 2-(acetylamino)-2-deoxy; 2: Betaine; 3: Isothiocyanatoacetaldehyde dimethyl acetal; 4: Lysine; 5: Serine; 6: Valeraldehyde; 7: L(-)-Fucose, tetramethyl ether; 8: Theophylline, 7-(2-hydroxy-3-(N-(2-hydroxyethyl)-N-methylamino)propyl); 9: Citronellol; 10: Bergamotol; 11: Muscimol; 12: 2,3-Anhydro-D-galactosan; 13: 1-Heptanol; 14: L-Alanine, N-dimethylaminomethylene; 15: (-)-Norephedrine; 16: Desulfo-glucobrassicin; 17: 2(3H)-Furanone, dihydro-5-propyl- γ -Heptanolactone; 18: Succinic acid, 2-methylphenyl 2-(dimethylamino)ethyl ester; 19: Decanoic acid, methyl ester; 20: Tridecanoic acid, methyl ester; 21: Hexanoic acid, methyl ester; 22: Stearic acid hydrazide; 23: Heptacosanoic acid, methyl ester; 24: Pentanoic acid, 4-methyl-, methyl ester; 25: 6-Octadecenoic acid, methyl ester, (Z); 26: 11-Octadecenoic acid, methyl ester; 27: 13-Docosenoic acid, methyl ester; 28: Pentadecanoic acid, 14-methyl-, methyl ester; 29: Tetradecanoic acid, 12-methyl-, methyl ester; 30: 1-Hexanol; 31: 9,12-Octadecadienoic acid, methyl ester, (E,E); 32: Oleic Acid γ -9-Octadecenoic acid (Z); 33: Heneicosanoic acid, methyl ester; 34: cis-Inositol; 35: 1-Butanol, 3-methyl; 36: (+/-)-Lavandulol, chlorodi-fluoroacetate; 37: 2,6,10,14,18-Pentamethyl-2,6,10,14,18-eicosapentaene; 38: Malonic acid, dihydroxy-, diisobutyl ester).

Consumption of Fagaceae and Fabaceae seeds suggested that this population researched foodstuffs with high energetic value, which were more reachable and cheaper than animal ones. Indeed, several oak species were naturally distributed into the woods surrounding the settlement place (Abbate et al., 2009). Acorns, being rich in carbohydrates (starch, 31–55%), proteins (2–8%), unsaturated fats (0.7–9%), minerals and vitamins (Al-Rousan et al., 2013; Pignone and Laghetti, 2010;

Šáliková et al., 2011), were commonly employed to ennobles other cereal flours or prepare astringent and anti-diarrheal medicinal decoctions (Rakić et al., 2006), after elimination of tannins by thermal treatment. On the other hand, legumes were usually adopted in poor diet in substitution of animal flesh, since they are rich in proteins (Pearson, 1997).

Starch granules and other microremains (e.g. phytoliths, plant

fibres) confirmed the use of C₃ plants (e.g. *Triticum* sp.) but also C₄ species (e.g. *S. bicolor*). These findings were coherent with carpological remains discovered in other coeval archaeological sites (Bosi et al., 2011). Certainly, these crops, being easily fit for cultivation at Mediterranean latitudes, represented important sources of carbohydrates for rural communities. According to Baldoni et al. (2016), the population of Colonna primarily employed C₃ plants, except that T67 US188 specimen, whose isotopic values strongly suggested the intake of C₄ species. However, our study revealed starch granules of C₄ plants in more than one individual (Table 2). Indeed, a minimal consumption of C₄ plants could not influence the overall isotopic signatures of the samples.

The detection of *D. carota* and *P. nigrum* starches was coherent with literature data; indeed, carrots were common ingredients of Medieval pottages (Lang, 2007; Mazzanti and Bosi, 2011), while black pepper was popular in Italy since Roman period and less expensive than other spices (Livarda, 2011).

The high number of unidentified starch granules could be related to prolonged cooking modality in boiling water, a typical procedure of soup preparation.

Gymnosperm pollen grains, identified in the ancient samples, suggested the presence of these species in the studied area. Accidental inhalation of pollen-rich air (Radini et al., 2017) or intentional consumption of resins, aromatic spirits, pine nuts or juniper berries, as documented in literature (Bosi et al., 2009; Casas-Agustench et al., 2011; Pieroni, 2000; Rösch, 2008), could be associated to this finding.

GC–MS method, applied for the first time on Medieval dental calculus, provided supplementary information which clearly defined the basal diet of the samples. In particular, markers of oil-rich seeds (e.g. pine nuts) and fruits and aquatic resources (e.g. fish, molluscs, algae) were found. Although in Baldoni et al. (2016) a fish-based diet was not linked to this population, our evidence allowed us to hypothesize a minimal or occasional utilization of fresh- and sea-water foods, probably due to its proximity to Nemi and Albano lakes, Tyrrhenian Sea and Rome.

Our data confirmed the typical Medieval consumption of Brassicaceae (e.g. cabbage, cauliflower) (Montanari, 2011, 2014). Moreover, chemical compounds typical of animal foodstuffs, Rosaceae fruits, grapes or wine and herbs (e.g. sage, rosemary, basil, mint, oregano) were also detected. In particular, these latter plant species were considered as taste regulators, salt substitutes, food preservatives and natural drugs (Cragg and Newman, 2013; Franzak and Pitta, 2005).

The use of tea, a species native of South-Eastern Asia, suggested the existence of commerce and trades with the Near East (Hoekman-Sites, 2007; Sharangi, 2009). Probably, the inhabitants of Colonna chewed tea leaves or used them in infusion as beverage (Wolf et al., 2008), due to their diuretic and exciting effect (Ferrara et al., 2001).

Surprisingly, the dental calculus of a specimen contained a fungal toxic molecule, the muscimol. It allowed us to elaborate two suppositions: i) the knowledge about mushrooms; indeed, although containing this poisonous alkaloid, some *Amanita* species are excellent after cooking (e.g. *A. solitaria*) (Weiß et al., 1998); ii) the possible medicinal purpose of this metabolite which, in low doses, miming GABA neurotransmitter, possesses analgesic properties (Molitoris, 1994; Wink, 1998). This last hypothesis was surely more adequate to the context, considering that the inhumate had a fracture of the right fibula, which could cause strong pains (Baldoni et al., 2016).

The therapeutic application of plant extracts was well known since antiquity (Gurib-Fakim, 2006); in fact, even today, vegetal tissues are easily available sources of pharmacological molecules. For this reason, the detection of natural bioactive compounds in the samples sustained the traditional use of wild medicinal plants in the community of Colonna.

Moreover, the identification of specific chemical indicators suggested the pharmaceutical application of several medicinal plants: *Digitalis* sp., maybe used as treatment of heart failure (Rastogi et al., 2016); *H. officinalis*, for its antiseptic and expectorant properties (Kizil

et al., 2010), *Artemisia* sp., showing digestive, antiseptic, antimalaric and expectorant properties (Karabegović et al., 2011); and *Ephedra* sp., a bronchodilator and vasoconstrictor species (Limberger et al., 2013). The use of these natural drugs in the studied population may be supported by the scientific evidence of skeletal degenerative pathologies, fractures and inflammations documented in Baldoni et al. (2016).

In conclusion, the present work, clarifying food habits of Colonna Medieval community, amplified modern knowledge about past environment, trade, plant medicinal uses and customs.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jasrep.2018.05.023>.

Conflict of interest

The authors declare no conflict of interest.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

- Abbate, G., Bonacquisti, S., Giovi, E., Iamónico, D., Iberite, M., Lorenzetti, R., 2009. Contribution to the vascular flora of the Castelli Romani Regional Park (Rome, Central Italy) with recent observations and early herbarium surveys. *Webbia* 64, 47–74.
- Al-Rousan, W.M., Ajo, R.Y., Al-Ismaïl, K.M., Attlee, A., Shaker, R.R., Osaili, T.M., 2013. Characterization of acorn fruit oils extracted from selected Mediterranean *Quercus* species. *Grasas Aceites* 64, 554–560.
- American Dental Association, 1999. Current Dental Terminology, third ed. CDT-3.
- Badler, V.R., McGovern, P.E., Michel, R.H., 1990. Drink and be merry! Infrared spectroscopy and ancient Near Eastern wine. In: *MASCA Research Papers in Science and Archaeology*. 7. pp. 25–36.
- Baldoni, M., Nardi, A., Muldner, G., Lelli, R., Gnes, M., Ferraresi, F., Meloni, V., Cerino, P., Greco, S., Manenti, G., Angle, M., Rickards, O., Martínez-Labarga, C., 2016. Archaeobiological reconstruction of the Italian medieval population of Colonna (8th–10th centuries CE). *J. Archaeol. Sci. Rep.* 10, 483–494.
- Bosi, G., Mercuri, A.M., Guarnieri, C., Mazzanti, M.B., 2009. Luxury food and ornamental plants at the 15th century AD Renaissance court of the Este family (Ferrara, northern Italy). *Veg. Hist. Archaeobotany* 18, 389–402.
- Bosi, G., Mazzanti, M.B., Florenzano, A., N'Siala, I.M., Pederzoli, A., Rinaldi, R., Torri, P., Mercuri, A.M., 2011. Seeds/fruits, pollen and parasite remains as evidence of site function: Piazza Garibaldi–Parma (N Italy) in Roman and Mediaeval times. *J. Archaeol. Sci.* 38, 1621–1633.
- Buckley, S.A., Stott, A.W., Evershed, R.P., 1999. Studies of organic residues from ancient Egyptian mummies using high temperature-gas chromatography-mass spectrometry and sequential thermal desorption-gas chromatography-mass spectrometry and pyrolysis-gas chromatography-mass spectrometry. *Analyst* 124, 443–452.
- Buckley, S., Usai, D., Jakob, T., Radini, A., Hardy, K., 2014. Dental calculus reveals unique insights into food items, cooking and plant processing in Prehistoric Central Sudan. *PLoS One* 9, e100808.
- Casas-Agustench, P., Salas-Huetos, A., Salas-Salvadó, J., 2011. Mediterranean nuts: origins, ancient medicinal benefits and symbolism. *Public Health Nutr.* 14, 2296–2301.
- Cragg, G.M., Newman, D.J., 2013. Natural products: a continuing source of novel drug leads. *Biochim. Biophys. Acta Gen. Subj.* 1830, 3670–3695.
- Cristiani, E., Radini, A., Edinborough, M., Boric, D., 2016. Dental calculus reveals Mesolithic foragers in the Balkans consumed domesticated plant foods. *Proc. Natl. Acad. Sci.* 113, 10298–10303.
- Crowther, A., 2012. The differential survival of native starch during cooking and implications for archaeological analyses: a review. *Archaeol. Anthropol. Sci.* 4, 221–235.
- Cummings, L.S., Yost, C., Soltysiak, A., 2016. Plant microfossils in human dental calculus from Nemrik 9, a Pre-Pottery Neolithic site in Northern Iraq. *Archaeol. Anthropol. Sci.* 1–9.
- Di Fiore, B., Chianese, D., 2008. Electric and magnetic tomographic approach for the geophysical investigation of an unexplored area in the archaeological site of Pompeii (Southern Italy). *J. Archaeol. Sci.* 35, 14–25.
- Dove, C.J., Koch, S.L., 2011. Microscopy of feathers: a practical guide for forensic feather identification. *Microscope-Chicago* 59, 51.
- Eglinton, G., Hamilton, R.J., Raphael, R.A., Gonzalez, A.G., 1962. Hydrocarbon constituents of the wax coatings of plant leaves: a taxonomic survey. *Nature* 193, 739–742.
- Ferrara, L., Montesano, D., Senatore, A., 2001. The distribution of minerals and flavonoids in the tea plant (*Camellia sinensis*). *Il Farmaco* 56, 397–401.
- FoodDB, 2013. FoodDB version 1.0. <http://fooddb.ca/>.
- Franzak, F., Pitta, D., 2005. New product development at eastern spice and flavorings. *J. Prod. Brand. Manag.* 14, 462–467.
- Ginatempo, M., 1984. Per la storia degli ecosistemi e dell'alimentazione medievale:

- recenti studi di archeozoologia in Italia. *Archeologia medievale* XI 35–61.
- Giovannini, F., 1994. Funzioni delle forme ceramiche e modelli alimentari medievali. In: De, Minicis (Ed.), *Le ceramiche di Roma e del Lazio in età medievale e moderna*. Roma, pp. 14–22.
- Gismondi, A., Di Marco, G., Martini, F., Sarti, L., Crespan, M., Martínez-Labarga, C., Rickards, O., Canini, A., 2016. Grapevine carpological remains revealed the existence of a Neolithic domesticated *Vitis vinifera* L. specimen containing ancient DNA partially preserved in modern ecotypes. *J. Archaeol. Sci.* 69, 75–84.
- Gismondi, A., D'Agostino, A., Canuti, L., Di Marco, G., Basoli, F., Canini, A., 2018a. Starch granules: a data collection of 40 food species. *Plant Biosystems*. <http://dx.doi.org/10.1080/11263504.2018.1473523>.
- Gismondi, A., Canuti, L., Rocco, G., Pisani, M., Ghelli, A., Bonanno, M., Canini, A., 2018b. GC-MS detection of plant pigments and metabolites in Roman Julio-Claudian wall paintings. *Phytochem. Lett.* 25, 47–51.
- Guasch-Jané, M.R., Ibern-Gómez, M., Andrés-Lacueva, C., Jáuregui, O., Lamuela-Raventós, R.M., 2004. Liquid chromatography with mass spectrometry in tandem mode applied for the identification of wine markers in residues from ancient Egyptian vessels. *Anal. Chem.* 76, 1672–1677.
- Gurib-Fakim, A., 2006. Medicinal plants: traditions of yesterday and drugs of tomorrow. *Mol. Asp. Med.* 27, 1–93.
- Hall, A.H., Hall, P.K., 1994. Ibotenic acid/muscimol-containing mushrooms. In: Spoeke, D.G., Rumack, B.H. (Eds.), *Handbook of Mushroom Poisoning, Diagnosis and Treatment*. CRC Press, Boca Raton, FL, pp. 265–278.
- Hardy, K., Blakeney, T., Copeland, L., Kirkham, J., Wrangham, R., Collins, M., 2009. Starch granules, dental calculus and new perspectives on ancient diet. *J. Archaeol. Sci.* 36, 248–255.
- Hardy, K., Buckley, S., Collins, M.J., Estalrich, A., Brothwell, D., Copeland, L., García-Tabernero, A., García-Vargas, S., de la Rasilla, M., Lalueza-Fox, C., Huguet, R., Bastir, M., Santamaría, D., Madella, M., Wilson, J., Cortés, A.F., Rosas, A., 2012. Neanderthal medics? Evidence for food, cooking, and medicinal plants entrapped in dental calculus. *Naturwissenschaften* 99 (8), 617–626.
- Hardy, K., Radini, A., Buckley, S., Sarig, R., Copeland, L., Gopher, A., Barkai, R., 2016. Dental calculus reveals potential respiratory irritants and ingestion of essential plant-based nutrients at Lower Palaeolithic Qesem Cave Israel. *Quat. Int.* 398, 129–135.
- Henry, A.G., Brooks, A.S., Piperno, D.R., 2011. Microfossils in calculus demonstrate consumption of plants and cooked foods in Neanderthal diets (Shanidar III, Iraq; Spy I and II, Belgium). *Proc. Natl. Acad. Sci.* 108, 486–491.
- Henry, A.G., Brooks, A.S., Piperno, D.R., 2014. Plant foods and the dietary ecology of Neanderthals and early modern humans. *J. Hum. Evol.* 69, 44–54.
- Hoekman-Sites, H.A., 2007. Using Residue Analysis to Confirm Trade Connections at Pella, Jordan. *Bar International Series* 1650, pp. 138.
- Kanthilatha, N., Boyd, W., Dowell, A., Mann, A., Chang, N., Wohlmuth, H., Parr, J., 2014. Identification of preserved fatty acids in archaeological floor sediments from prehistoric sites at Ban Non Wat and Nong Hua Raet in northeast Thailand using gas chromatography. *J. Archaeol. Sci.* 46, 353–362.
- Karabegović, I., Nikolova, M., Veličković, D., Stojičević, S., Veljković, V., Lazić, M., 2011. Comparison of antioxidant and antimicrobial activities of methanolic extracts of the *Artemisia* sp. recovered by different extraction techniques. *Chin. J. Chem. Eng.* 19, 504–511.
- Kizil, S., Hasimi, N., Tolan, V., Kiliç, E., Karataş, H., 2010. Chemical composition, antimicrobial and antioxidant activities of hyssop (*Hyssopus officinalis* L.) essential oil. *Notulae Botanicae Horti Agrobotanici Cluj-Napoca* 38, 99.
- Kris-Etherton, P.M., Harris, W.S., Appel, L.J., 2003. Fish consumption, fish oil, omega-3 fatty acids, and cardiovascular disease. *Arterioscler. Thromb. Vasc. Biol.* 23, 20–30.
- Lang, P., 2007. In: Hartmann, Sieglind (Ed.), *Fauna and Flora in the Middle Ages*. Peter Lang GmbH, pp. 170–194.
- Leonard, C., Vashro, L., O'Connell, J.F., Henry, A.G., 2015. Plant microremains in dental calculus as a record of plant consumption: a test with Tve forager-horticulturalists. *J. Archaeol. Sci. Rep.* 2, 449–457.
- Lieverse, A.R., 1999. Diet and the aetiology of dental calculus. *Int. J. Osteoarchaeol.* 9, 219–232.
- Limberger, R.P., Jacques, A.L.B., Schmitt, G.C., Arbo, M.D., 2013. Pharmacological effects of ephedrine. In: *Natural Products*. Springer Berlin Heidelberg, pp. 1217–1237.
- Lindhe, J., 1984. *Parodontologia*. Edi. Ermes, Milano, pp. 110–117.
- Livarda, A., 2011. Spicing up life in northwestern Europe: exotic food plant imports in the Roman and medieval world. *Veg. Hist. Archaeobotany* 20, 143–164.
- Mazzanti, M.B., Bosi, G., 2011. Informazioni etnobotaniche dai rifiuti della Ferrara medievale-rinascimentale. In: Bonini, G., Brusa, A., Cervi, R., Garimberti, E. (Eds.), *Il paesaggio agrario italiano medievale*. Quaderni 7. Istituto Alcide Cervi, pp. 99–109.
- Molitoris, H.P., 1994. Mushrooms in medicine. *Folia Microbiol.* 39, 91.
- Montanari, M., 1979. *L'alimentazione contadina nell'alto Medioevo*. Liguori Publications, Napoli.
- Montanari, M., 1988. *Alimentazione e cultura nel Medioevo*. Laterza, Roma-Bari.
- Montanari, M., 2011. *Il cibo come cultura*. Laterza & Figli Spa.
- Montanari, M., 2014. *Gusti del Medioevo: i prodotti, la cucina, la tavola*. Gius. Laterza & Figli Spa.
- Montanari, M., Brombert, B.A., 2015. *Medieval Tastes: Food, Cooking, and the Table*. Columbia University Press, New York.
- Müldner, G., Richards, M.P., 2005. Fast or feast: reconstructing diet in later medieval England by stable isotope analysis. *J. Archaeol. Sci.* 32, 39–48.
- Pearson, K.L., 1997. Nutrition and the early-medieval diet. *Speculum* 72, 1–32.
- Piccinni, G., 2002. La campagna e la città (secoli XII–XV). In: Cortonesi, A., Pasquali, G., Piccinni, G. (Eds.), *Uomini e campagne nell'Italia medievale*. Laterza, Roma-Bari, pp. 125–189.
- Pieroni, A., 2000. Medicinal plants and food medicines in the folk traditions of the upper Lucca Province, Italy. *J. Ethnopharmacol.* 70, 235–273.
- Pignone, D., Laghetti, G., 2010. On sweet acorn (*Quercus* spp.) cake tradition in Italian cultural and ethnic islands. *Genet. Resour. Crop. Evol.* 57, 1261–1266.
- Radini, A., Nikita, E., Shillito, L.M., 2016. Human Dental Calculus and a Medieval Urban Environment. *Everyday Life in Medieval Europe: Environmental and Artefactual Approaches to Dwelling in Town and Country*. Brepols, Belgium (<http://dx.doi.org/10.1073/pnas.1018116108>).
- Radini, A., Nikita, E., Buckley, S., Copeland, L., Hardy, K., 2017. Beyond food: the multiple pathways for inclusion of materials into ancient dental calculus. *Am. J. Phys. Anthropol.* 162, 71–83.
- Rakić, S., Povrenović, D., Tešević, V., Simić, M., Maletić, R., 2006. Oak acorn, polyphenols and antioxidant activity in functional food. *J. Food Eng.* 74, 416–423.
- Rastogi, S., Pandey, M.M., Rawat, A.K.S., 2016. Traditional herbs: a remedy for cardiovascular disorders. *Phytomedicine* 23, 1082–1089.
- Rösch, M., 2008. New aspects of agriculture and diet of the early medieval period in central Europe: waterlogged plant material from sites in south-western Germany. *Veg. Hist. Archaeobotany* 17, 225–238.
- Salamon, M., Coppa, A., McCormick, M., Rubini, M., Vargiu, R., Tuross, N., 2008. The confluence of historical and isotopic approaches in reconstructing the medieval Mediterranean diet. *J. Archaeol. Sci.* 35, 1667–1672.
- Šálková, T., Divišová, M., Kadochová, Š., Beneš, J., Delawská, K., Kadlčková, E., Němečková, L., Pokorná, K., Voska, V., Žemličková, A., 2011. Acorns as a food resource. An experiment with acorn preparation and taste. *Interdisciplinaria Archaeologica, Natural Sciences in Archaeology* 2, 139–147.
- Salvadori, F., 2003. Archeozoologia e Medioevo: lo stato degli studi. In: Fiorillo, R., Perduto, P. (Eds.), *III Congresso Nazionale di Archeologia Medievale*. Firenze, pp. 176–181.
- Salvini, L., Pecci, A., Giorgi, G., 2008. Cooking activities during the Middle Ages: organic residues in ceramic vessels from the Sant'Antimo Church (Piombino-Central Italy). *J. Mass Spectrom.* 43, 108–115.
- Sharangi, A.B., 2009. Medicinal and therapeutic potentialities of tea (*Camellia sinensis* L.) – a review. *Food Res. Int.* 42, 529–535.
- Swanson, D., Block, R., Mousa, S.A., 2012. Omega-3 fatty acids EPA and DHA: health benefits throughout life. *Adv. Nutr.* 3, 1–7.
- TGSC, 2015. *The Good Scents Company*. <http://www.thegoodscentscompany.com/>.
- Travaglini, A., Arseni, A., Brighetti, M.A., Vinciguerra, F., 2014. *Atlante del polline delle principali specie allergeniche d'Italia*. (Il Raggio Verde).
- Tucci, H.Z., 1985. Il mondo medievale dei pesci tra realtà e immaginazione, in: *L'uomo di fronte al mondo animale nell'alto medioevo*, XXXI Settimana di Studio del Centro Italiano di Studi sull'Alto Medioevo (7–13 aprile 1983). CISAM, Spoleto, pp. 291–360.
- Tuliani, M., 2001. *La tavola imbandita: note su alimentazione e società a Siena nel Medioevo*. Betti, Siena.
- Weiß, M., Yang, Z.L., Oberwinkler, F., 1998. Molecular phylogenetic studies in the genus *Amanita*. *Can. J. Bot.* 76, 1170–1179.
- Weyrich, L.S., Duchene, S., Soubrier, J., Arriola, L., Llamas, B., Breen, J., Morris, A.G., Alt, K.W., Caramelli, D., Dresely, V., Farrell, M., Farrer, A.G., Francken, M., Gully, N., Haak, W., Hardy, K., Harvati, K., Held, P., Holmes, E.C., Kaidonis, J., Lalueza-Fox, C., de la Rasilla, M., Rosas, A., Semal, P., Soltysiak, A., Townsend, G., Usai, D., Wahl, J., Huson, D.H., Dobney, K., Alan Cooper, A., 2017. Neanderthal behaviour, diet, and disease inferred from ancient DNA in dental calculus. *Nature* 544, 357–361.
- Wink, M., 1998. Modes of action of alkaloids. In: *Alkaloids*. Springer US, pp. 301–326.
- Wolf, A., Bray, G.A., Popkin, B.M., 2008. A short history of beverages and how our body treats them. *Obes. Rev.* 9, 151–164.